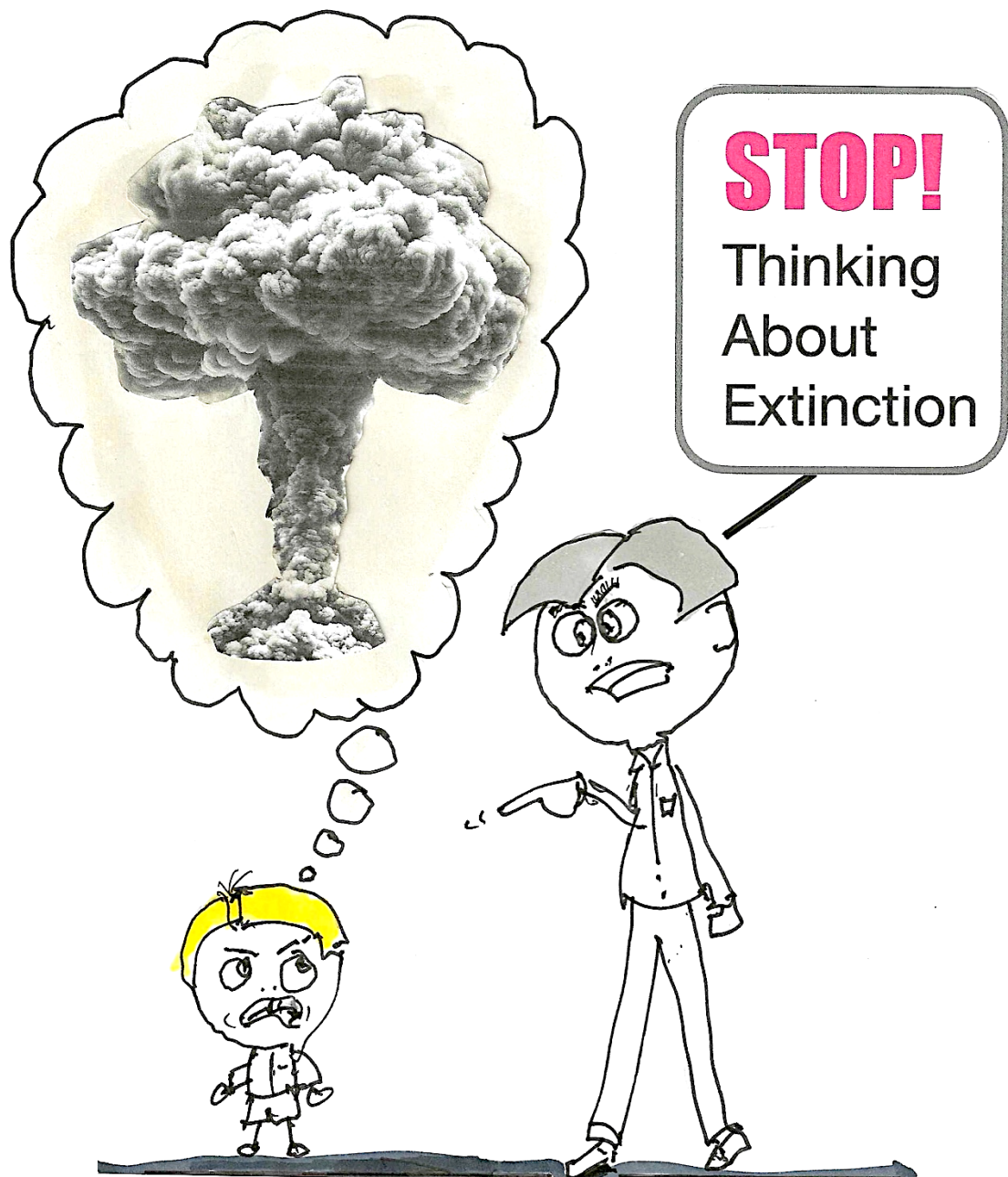




From the Atomic Bomb to A.I.
My 50 Years of **Anxiety**
David Rejeski

"If there was an observer on Mars, they would probably be amazed that we have survived this long. There are two problems for our species' survival - nuclear war and environmental catastrophe - and we're hurtling towards them. Knowingly. This hypothetical Martian would probably conclude that human beings were an evolutionary error."

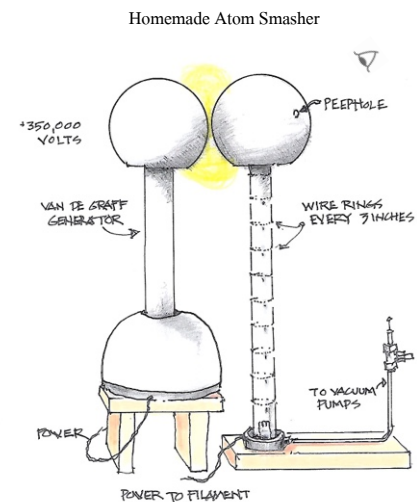
— Noam Chomsky¹

"I would estimate the chances are about 49 percent that the world as we know it will collapse by about 2050."

- Jared Diamond²

When I was a kid growing up in the 1950s, extinction was part of my life. I didn't obsess about my demise. There were after all diversions like the 1957 World Series when the Milwaukee Braves beat the Yankees, but the reminders of planetary disaster were omnipresent: my mother's incessant stockpiling of canned goods in the corner of our cellar, weekly air raid drills at my elementary school, and the hand-made Geiger counter (my sixth-grade science project) that stood ready on the shelf for use 'after' the bomb went off. In retrospect, extinction is not something a normal 8-year-old should be worried about, but one single technology — the atomic bomb — changed that forever. Studies conducted in the decades following the Cold War showed that children exposed to the constant threat of global extinction were more likely to suffer from anxiety disorders.³ I did, later in life, though causality is difficult to prove. But looking back on my childhood, I was too busy thinking about technology's wonders to worry about its downsides.

Each month, the latest edition of *Scientific American* would be sticking out of our rusting mailbox. Each issue had a section entitled "The Amateur Scientist" and there were projects on these pages that would absolutely terrify today's helicopter parents and unleash waves of litigation across large swaths of America's suburbs. Included were plans to build a six-foot long, solid fuel rocket that could climb to over 2,000 feet; a home-made atom smasher (illustration right); an inexpensive X-ray machine; or your very own seismological observatory.⁴ I bought into all of this — every volt, hydrogen bond, approaching comet, puzzle solving machine, and satellite tracker. In 2001, *Scientific American* stopped publishing this section after a run of 72 years, but I hoped and patiently waited for a new resurgence in amateur science. Someday it would happen, and I would be part of it again.



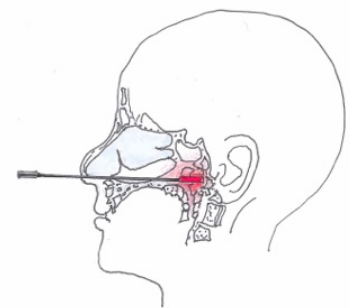
One tactic I learned early in life was to focus on the good emerging from these technologies, not the bad. That was infinitely easier for kids my age. We now know from neuroscience research that children "...are hyper-optimistic, because they learn much less from bad outcomes. Whilst children learn as much as adolescents when something good happens, they tend to ignore when things are not as good as they hoped."⁵ I simply wanted to believe. Technology was like road kill to me, I could not look away. And reinforcement appeared around every corner as I watched my black-and-white television morph into a kaleidoscope of colors, cars grew tail fins of enormous amphibious proportions, and three-course meals

magically appeared in neatly partitioned aluminum trays with packaging that touted: “What a meal! A homestyle touch.”⁶

And then there was Sputnik, launched on October 7, 1957. A gleaming sphere visible in the night sky that embodied everything dear to me: rocket motors, aerodynamic forces, electronics, optics, radio beacons, and promises of future visits to the moon, planets, even other galaxies. The amount of consternation that Sputnik caused in military and intelligence circles cannot be underestimated, and has become clearer since the declassification of documents from that era. After Sputnik was launched, the U.S. put planes with hydrogen bombs in the air around the clock and Edward Teller, the so-called ‘father of the hydrogen bomb,’ went on television to proclaim the U.S. had “lost a battle more important and greater than Pearl Harbor.”⁷ But for kids, the bright, beeping sphere was not a threat to American notions of Cold War scientific supremacy but a bridge to the extraterrestrial outposts and galactic adventures that were omnipresent in our daily diet of Marvel comics. Comics were our refuge, the bane of our parents; our social media.⁸

But the fantasies, the comics, the late nights in front of my shortwave radio, the flights of fancy were also a mental filter that blocked the world outside my neatly arranged suburban life. In the year after Sputnik, the United States detonated 77 above-ground atomic bombs, many in the American southwest, a record exceeded only in 1962.”⁹ A few decades later, millions of ordinary U.S. citizens, like me, would get the bill for our scientific exuberance and drive for geopolitical one-upmanship during the Cold War era. Elevated levels of radioactive iodine were discovered in cow’s milk 3,000 miles from the Nevada testing grounds and in thyroid glands of milk-drinking kids like me. In 1998 the Congress asked the National Cancer Institute to estimate the health impacts of nuclear weapons tested in the atmosphere.¹⁰ Running my data through their on-line calculator told me that my risk of developing thyroid cancer was twice that of the unexposed.¹¹ But I am still alive.

Sometime in 1957, I was sitting in the doctor’s office — again — after another cold with accompanying earache had left me unable to hear. A small wooden cart was dragged clanking into the room with two silver, and obviously heavy, cylindrical canisters with protruding rods and I watched in amazement as each thin, six-inch rod was guided deeply into my nasal cavities. This was my first close encounter with the atom — that invisible particle that I had gotten to know in *Captain Marvel Meets Mr. Atom* (Marvel Comic #78). Each rod was tipped with a capsule of radioactive radium, now sitting millimeters away from my pituitary and thyroid glands, each promising a quick, painless cure for my auditory handicaps. I wanted to believe, but nothing happened. Only later would I learn that I had been exposed to the radiation equivalent of around 10,000 dental X-rays.¹²



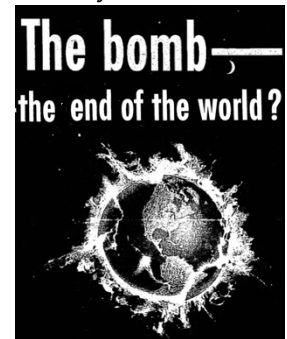
I became one of an estimated 500,000 to 2.5 million Americans that were subjected to nasopharyngeal radium irradiation — NRI for short — most procedures conducted in the comfort of your doctor’s office.¹³ One early patient later referred to the procedure as “Nagasaki

up the nose.”¹⁴ The company that supplied the radium, the Radium Chemical Company, located in Queens, New York, was shut down in 1981. After it was demolished, the Environmental Protection Agency found 10,000 radium needles, 92 tons and 197 gallons of nonradioactive hazardous waste, and 812 tons of radioactive soil and debris.¹⁵ A jaw-dropping inventory of historical items containing radium was completed in 2008 under contract to the Nuclear Regulatory Commission and included a wide range of ‘quack’ cures such as Dormad Radioactive Toothpaste, sure to “increase the defenses of the teeth and gums;” consumer products like condoms, beer, refrigerator deodorizer, and glow-in the dark costumes; and for the kids, radium-based toys such as the Tom Mix Magic-Light Tiger-Eye Ring and the Gilbert U-238 Atomic Energy Lab.¹⁶ Such wonders!

Angela Creager, a Princeton University historian, who has studied the use of radioactive substances in medicine during the 1950s, observed that we “... think the government should be involved at some level in protecting civilians from exploitation by a medical establishment that is eager for research subjects, but in the 1950s the thought was that physicians should be trusted to regulate themselves.”¹⁷ We tend to forget that these were the same experts (always male) in white coats who ensured the public that there was “Not one single case of throat irritation due to smoking Camels,” or Lucky Strikes, or Viceroyes, or Chesterfields, for that matter.¹⁸ Most believed all of this, and more. And why not? In the early 1960’s trust in the US medical profession had reached almost 75 percent (now below one third), and during the Eisenhower administration, trust in government was around 75 percent according to the National Election Study (now below one third and dropping).¹⁹ As humans, we are wired to trust, and whether it is doctors taking money from drug companies or climate scientists in bed with the fossil fuel industry, we tend to overlook often blatant conflicts of interest that undermine the latest claim, report, or news story.²⁰

Trust bends the truth in different directions and technological risks and benefits are refracted through the prism of trust. But in my early years, there was no inner alarm that warned me that my trust was somehow misplaced or manipulated, and no moral reckoning with the adult world was possible. That would come later. Though I did not sense it then, I was living through an inflection point, difficult to pinpoint exactly in time, but probably occurring sometime in the decade after 1945, and the effects would be long-lasting. Anthropologist Joseph Mosco has noted that, “America in the twenty-first century remains a society built around, and to a large extent, through the bomb.”²¹ This legacy is not just about technology, in fact, it has little to do with technology as commonly defined, but about the institutions, narratives, expectations, and stories that surround our technologies and technological choices.

The first shift I became aware of as I grew older was being confronted with the scepter of human extinction at a planetary scale. Exactly how life would be snuffed out remained a matter of conjecture, but one scenario involved the earth being vaporized by a chain reaction set off by a thermonuclear explosion that literally ignited the atmosphere.²² Discussions of this scenario took place among physicists before the atomic bomb was detonated in New Mexico, and again before the first thermonuclear device as exploded at Bikini Atoll in the Pacific Ocean. The scenario was even presented to Hitler who, “...was plainly not delighted with the possibility that the earth under his rule might be transformed into a glowing star. Occasionally, however, he joked that the scientists in their unworldly urge to lay bare all the secrets under heaven might someday set the globe on fire.”²³ The possibility that a thermonuclear explosion could result in the end of the planet remained largely below the surface of public awareness until the publication of a 1959 essay by the writer and novelist Pearl S. Buck in the popular Sunday magazine supplement, *The American Weekly*, entitled “The Bomb — The End of the World?”²⁴ But subsequent bomb tests, with yields up to 1,500 times more powerful than the Hiroshima bomb, failed to ignite the earth’s atmosphere.²⁵ As with many risks we have come to face as a species, catastrophic outcomes with low probabilities are often viewed in retrospect as overblown (if they don’t outright kill us), and those who ring alarm bells labeled as fear mongers, Luddites, or worse. Another common extinction theory involved nuclear winter, a global cooling caused by fine dust particles rising from multiple, high-yield nuclear surface bursts, which would effectively block out the sun.²⁶ Freezing to death, of course, was the fate of those not killed directly by blast shock or radiation effects from the nuclear explosions.



This transition to a world shaped by you, and I, and our human brethren now has a name, coined in 2000, the *Anthropocene*, and a group of scientists —the Anthropocene Working Group —has been meeting since 2009 to determine exactly when humanity crossed the boundary.²⁷ Recently, in a 29-to-4 vote, they recommended that the Anthropocene’s beginning be set in the mid-20th century — my childhood.²⁸ The signals of this transition can be seen geologically at multiple sites across the planet in the form of radioactive isotopes scattered from nuclear tests, ash from fossil fuel combustion, and long-lasting residues from the widening use of synthetic chemicals, including fertilizers, pesticides, and plastics.²⁹ Implicit in this new state of affairs is that humans have taken significant control over the chemical and biological cycles on which the earth and its species depend. The carbon cycle, the nitrogen cycle, and parts of the hydrological cycle are today all firmly in human hands (and, as some have postulated, may be shortly in the hands of algorithms, which we may, or may not, control).

Post-1950, other shifts occurred, less overtly threatening as human extinction, or less impressive as the emergence of a new geological era, but these would penetrate deeper into the human psyche and social fabric. Within a few short decades, we saw the emergence of what Yale sociologist Kai Erikson termed ‘a new species of trouble.’³⁰ Erikson had spent years studying human responses to disasters and accidents from dam breaks to toxic releases and noticed that communities reacted differently to disasters caused by human hands, rather than what are often perceived as ‘acts of God.’ For instance, pervasive, invisible, and incomprehensible toxins evoke long lasting anxieties and uncertainties not just in individuals, but their families, friends, communities, and progeny. People become and remained fearful and

perplexed, searching for answers and causality. One day after the atomic bomb exploded over Hiroshima, Michihiko Hachiya, a surviving doctor wrote in his diary: "What kind of bomb was it that had destroyed Hiroshima? Whatever it was, it did not make sense." "Whatever it was, it was beyond my comprehension."³¹ Importantly, these threats violate our notion of plot, stories that have a beginning, middle, and end — a final chapter where we rebuild, and life resumes its normal rhythms. As Erikson noted "An all clear is never sounded. The book of accounts is never closed."

These new species of trouble have never stopped their advance, multiplying in number and kind. Think of today's so-called 'forever chemicals,' persistent toxics that have found their way into hundreds of consumer products, packaging, our soil, our water, and our foods.³² Each successive technological breakthrough has been accompanied by its own set of hopes and fears, and the downsides of scientific advances have often been foreseen and then exploited in the parallel universe of science fiction. In Don DeLillo's 1985 sci-fi classic, *White Noise*, now a Netflix movie starring Adam Driver and Greta Gerwig, a cloud of Nyodene Derivative engulfs a sleepy mid-western town after a train accident, triggering panic and a massive evacuation. One character in the novel notes that, "The greater the scientific advance, the more primitive the fear." The recent train derailment and chemical release in East Palestine, Ohio, bears an uncanny similarity to the scenario that plays out in DeLillo's novel, reflecting the fluid boundary between fiction and non-fiction.³³ An analysis of data from the Environmental Protection Agency and civil society groups indicated that accidental releases of chemicals in the U.S. are now occurring once every two days.³⁴

In Michael Crichton's 2002 novel *Prey* about runaway nanobots from a secret Nevada lab that threaten to engulf the earth in grey goo, the characters confront recurring ethical and scientific challenges at the intersection of nanotechnology, biotech, and computer science (think programmable, invisible, and reproducing machines). In the words of *Prey*'s protagonist, "Things never turn out the way you think they will." The law of unintended consequences is never vanquished.³⁵

To their credit, some scientists understand the public responses and ambivalence that can arise from novel toxic threats, often invisible, with impacts stretching years or decades into the future. Leon Kass, the former chairman of the President's council on bioethics, has described this public reaction as the *Yuck Factor* — 'an emotional expression of deep wisdom, beyond reason's power fully to articulate it.'³⁶ In many cases, public perceptions and acceptance (yuck or no yuck) depend on context, specifically how a technology will be applied, the safeguards surrounding its use, and overall trust in those who create and potentially regulate the technology. The mental transition that takes people from 'cures for cancer' to 'causes for concern' and back, has affected everything from nanotechnology to gene-edited babies and burgers from bio-reactors.³⁷

And, of course, the old threats can reappear, as psychiatrist Robert Jay Lifton recently noted: "Nuclear fear has become a model for large threats that endanger the human future. So when people encounter climate change, or now COVID-19, they find themselves associating back to nuclear images."³⁸ A recent Reuters/Ipsos poll surveying reactions to the Ukraine crisis found that 58% of Americans are afraid we are headed toward nuclear war with Russia.³⁹ "The

level of anxiety is something that I haven't seen since the Cuban missile crisis," said Peter Kuznick, a history professor and director of the Nuclear Studies Institute at American University.⁴⁰ Even the concept of 'nuclear winter' has reappeared in the press as a possible consequence of a nuclear conflagration.⁴¹ Or is it a solution? Complicating our ability to separate problems from solutions is the idea that floating fine particles in the upper atmosphere could actually help combat climate related warming, a highly controversial approach involving solar radiation management, a subset of so-called geo-engineering strategies.⁴² Neal Stephenson's recent sci-fi thriller, *Termination Shock*, explores the powerful allure of an easy and relatively inexpensive solution to the climate crisis, and what might happen if the planetary cooling system is turned off.⁴³ Welcome to the Anthropocene.

With every downside, there seems to be an upside. Counterbalancing emerging threats and the possibility of extinction was the promise of progress and the almost irresistible pull of new knowledge. It is hard to say 'no' to science and its technological offspring. I couldn't as a kid. As philosopher Donna Haraway once said, "science is the greatest story ever told."⁴⁴

The generation of new knowledge — better organized, funded, and directed towards goals — brought with it an unquestioned, and almost unquestionable, inevitability. Robert Oppenheimer noted that the use of the atomic bomb, "was implicit in the project."⁴⁵ This attitude was echoed by Oppenheimer's colleague and fellow physicist Samuel Goudsmit: "If we have such a weapon, we are going to use it," and later by French anthropologist Jacques Soustelle: "Since it was possible, it was necessary."⁴⁶ President Roosevelt clearly grasped the possibilities of applying this forward momentum to a peacetime economy when he wrote in 1944 that, "New frontiers of the mind are before us, and if they are pioneered with the same vision, boldness, and drive with which we have waged this war we can create a fuller and more fruitful employment and a fuller and more fruitful life."⁴⁷ And the money flowed. Back in 1940, expenditures for scientific research by industry and government in the United States amounted to around \$309 million (over \$6.1 billion in today's dollars) but have now risen to over \$700 billion in investments by business, higher education, the federal government, nonprofit organizations, and others but mostly by the business sector.⁴⁸

In retrospect, investing in science was not a bad bet. Quantifying the link between R&D investment, innovation, and economic growth has vexed economists for decades, because the return on investments can vary radically by country, business sector, and technology, but studies indicate an average return of around 20 percent.⁴⁹ This ideas-driven growth model was tantalizing for policy makers, but already by the late 1950's, questions had arisen about what and who guided the intentions and goals of these investments, or, to what degree these technology trajectories could even be shaped — questions that we still grapple with today. And by the late 1960's, much of the sheen of Big Science had worn off, leading some to comment that, "the 20-year honeymoon for science is drawing to a close."⁵⁰

In 1955, the Dutch sociologist Fred Polak wrote a book exploring the power and role of images of the future, differentiated between "what must be" and "what ought to be," between inevitability and choice. He emphasized that, "The first point of view sees history as a book that has already been written; the second sees history as a process that man can or cannot manipulate."⁵¹ It's our choice, or is it? In the wake of the Manhattan Project, observers

questioned the very idea that humans could choose and shape technologies “in the direction of good and not of evil” and pondered what instruments could be used to accomplish such steering.⁵² Attempts were made to balance the bad with the good, with an eye towards reducing public concerns. After the atomic bomb came ‘atoms for peace’ with a long list of promises from ‘energy too cheap to measure’ to ‘cures for cancer.’⁵³ The atoms for peace program even received its own 3-cent stamp, issued in 1955. The atomic age ushered in another view of science, that would later be termed *dual use* — that good and evil were inseparable parts of the whole deal. Extinction may not be an unfortunate and unintended consequence of science, it could be, in fact, the goal of the scientific enterprise or the result of a misguided use of technology.⁵⁴

As we moved into the 20th century, some would note that a commonly held view was that, “technology is neither good nor bad...It depends on us and our values whether it is used well or ill. Whenever politicians or business people are confronted with contentious issues of technological culture, they invoke the instrumentalist reading of technology.”⁵⁵ Recently, former US Secretary of Defense Ashton Carter noted that, “Disruptive scientific and technological progress is not...inherently good or inherently evil. But its arc is for us to shape.”⁵⁶ Tim Cook, the CEO of Apple, echoed this sentiment when he said, “Technology doesn’t want to be good or bad, it’s in the hands of the creator whether it becomes good or bad.”⁵⁷ Some people can’t make up their minds. Sam Altman, the CEO of OpenAI, the company that developed ChatGPT, has criticized the recent over-hyped calls for a moratorium on AI development, while also urging caution and the creation of AI oversight systems.⁵⁸ Since everybody seems to be turning to ChatGPT for answers, I decided to ask AI. Here is what the chatbot told me about technology (after about two seconds of deliberation): “Technology itself is neutral, meaning it’s neither inherently good nor bad. It is simply a tool created by humans to serve a particular purpose. Whether technology has a positive or negative impact depends on how it’s used.” Of course, the bot is trained on a steady and growing diet of human thought and ideas so the response was hardly surprising. But is it worrisome?⁵⁹

Once science unlocks a new door to another future, must we walk through? Who creates the map and who provides the guardrails? And who is the ‘who’? How was ChatGPT unleashed on a largely unsuspecting world? The explanation is simple and sobering: “Decisions at the company [Open AI] are usually made by consensus: employees debate, experts are consulted, and eventually a joint conclusion is reached. But when the question of the chatbot’s release cycled up to [CEO] Sam Altman he said he made a rare “contentious unilateral decision.”⁶⁰ In an interview conducted in 2019, Altman sounded a lot like Robert Oppenheimer, stating that, “Technology happens because it is possible.”⁶¹ In April 2023, Dr. Geoffrey Hinton, often called “The Godfather of AI” left Google and more openly voiced his concerns about the potential impacts of AI applications like ChatGPT. Hinton admitted he used to quote from Oppenheimer when asked about his own involvement in the development of AI: “When you see something that is technically sweet, you go ahead and do it,” but Hinton does not quote Oppenheimer anymore.⁶²

Can we expect scientists and entrepreneurs to police themselves? What about companies engaged in winner-takes-all, competitive struggles to dominate markets? Rouge experimenters in garage labs? Should governments step in, what about non-governmental

organizations, trade organizations, or advocacy groups? And if interventions are to be made, can they be made in time to ensure better, more equitable, outcomes for humanity? Was Einstein right when he famously observed, “the man of science is a poor moral philosopher.”⁶³

In May 2019 I visited a cafe in trendy Kreuzberg Berlin for a meeting of Extinction Rebellion (abbreviated XR), a social and political activism group founded in the UK in October 2018 in response to the mounting evidence of global warning and reluctance of policy makers to address it. I wanted to find out what the ‘new’ extinction was all about. Was this extinction different from the one I had fretted about as a youngster — scarier; more inevitable? And why has this concept again emerged as a *leitbild* for humanity, with activists, the media, and even the United Nations urging the use of ever stronger language to describe our predicament and trigger more alarm bells: “it’s not climate change, it’s a climate breakdown, a crisis, an emergency!”⁶⁴

Extinction Rebellion spawned numerous off-shoots, including the Last Generation, a group of activists who have gained media coverage by gluing themselves to highways during rush hour; defiling art works, which included throwing a black liquid on Gustav’s Klimt’s painting *Death and Life* in the Leopold Museum in Vienna; and dumping black dye into the Barcaccia fountain at the base of the Spanish Steps in central Rome.⁶⁵ “*We are the seeds, we are the seeds. They thought they could bury us, they didn’t know, we are the seeds,*” was the refrain of a Last Generation march through Berlin in April 2023 that threatened to shut down the city. Criticisms of the group’s tactics by police, the general public, and even the head of Germany’s Justice Ministry, who compared them to protesters in Germany in the early 1930s, has resulted in less glue and a shift to blocking urban traffic by walking slow.⁶⁶ Some studies have shown that moderate groups of the larger climate movement, such as *Fridays for Future*, actually benefit from attacks by more radical factions (known as the radical flank effect).⁶⁷ The group’s tactics and public blowback have overwhelmed any discussion of their actual goals to address climate change. But the group’s use of the term ‘last’ seems ever-more relevant and resonant since an Australian study found that among a group of 10–14-year-olds about 27% believed that the world may end during their lifetime due to climate change and other global threats and a German study found that two-thirds think climate change threatens human existence.⁶⁸

Recently some observers have noted that these climate ‘doomers’ are now replacing the climate ‘rejectors’ and ‘deniers.’⁶⁹ All of this negativity has given rise to a counter movement, the Tech or Decarb Bros, a bunch of techno-optimists who speak out against ‘doomerism’ and believe that the magic formula of green tech + money will bail out the planet.⁷⁰ But, for the most part, it seems like worry, if not outright panic, is in order. At the close of the recent COP 28 climate conference UN Chief Antonio Guterres talked of an urgent need for a “how-to-guide to defuse the climate time bomb.”⁷¹ Even members of the scientific community, who normally try to avoid hyperbole, and have a tendency “to err on the side of least drama,” have pointed out that we are underestimating the chances of a ‘ghastly’ future and urged their colleagues to speak out more forcefully, saying that today’s situation places “extraordinary responsibility on scientists to speak out candidly and accurately when engaging with government, business, and the public.”⁷²

But there is a problem. Whatever we want to call the climate situation — a catastrophe, breakdown, emergency, or crisis — it's now accompanied by the water crisis, the food crisis, the biodiversity crisis, the micro plastics crisis, the COVID-19 crisis, the financial crisis, the general crisis in confidence, and the re-emerging threat of a nuclear exchange or a reactor meltdown in the Ukraine.⁷³ This layer cake of crises solicits a response that can numb us, which may diminish our fears, but often undercuts our ability to respond, either individually or collectively. When I was a child, I worried about the atomic bomb, maybe polio (but there was a vaccine, I learned), and having my supply to Hostess Twinkies cut off, but not much more. Most of us have a limited supply of worry to spread around and we must decide where to focus our emotional reserves — on a negative bank account statement, a possible long-COVID infection, or rising sea levels.⁷⁴ Worries combined with a steady diet of negativity and bad news can lead to paralysis and undercut the creativity needed to solve problems.

Recently, the term 'existential threat' (ET, for short, or the e-word) is trumping 'crisis' as the favorite go-to term of climate *doomers*, financial *doomers*, and pandemic *doomers*. Existential was [Dictionary.com](https://www.dictionary.com)'s Word of the Year in 2019.⁷⁵ But, as a recent opinion piece pointed out, "To label something a threat or a risk or a danger, without using the bombastic e-word as a verbal crutch, would encourage those who issue warnings to be specific and those who receive them to be deliberate. And no matter how serious an impending problem might be, freaking out does not make it easier to solve."⁷⁶

When climate activist Greta Thunberg addressed the World Economic Forum in 2019 she said, "Our house is on fire. I don't want your hope, I want you to panic."⁷⁷ But panic comes at a price. Researchers have found that, "when people are prompted with positive stimuli, they do better at creative tasks. They literally see the big picture."⁷⁸ Positive aspirations can drive personal achievement, while social and political engagement is often undermined by a sense of pessimism around the future.⁷⁹ In a recent study involving a large sample of almost 5,000 Americans, hope was a better predictor of climate action than political orientation or just feeling anxious.⁸⁰ But a 2022 survey found that, "About one in four Americans (27%) say they 'try not to think about global warming' and 15% say they 'try to avoid information about global warming.'"⁸¹

There is now a collapse of hope, with survey research "suggest[ing] a *hope gap* among the American public, despite the myriad efforts underway to address climate change at individual to international scales."⁸² I started looking for an expert on hope and I found one, Matthew Ballew, a social and environmental psychologist who worked on the study that identified the so-called hope gap. I asked Ballew how people in his studies viewed the role of technology in relationship to hope about the future. He indicated that reactions often reflect a simplistic view concerning the role of technology, that it will simply bail us out, with little or no effort on our part or the part of our leaders -- technology as wishful thinking: "...technology usually finds a way to emerge when we need it most," "...maybe [scientists] they can come up with ideas," or "technologies will improve that will make it more profitable to get clean energy [but] I'm really not very hopeful." The other view is more nuanced and posits technology as only part of the answer, requiring more investments, political will, leadership, and efficiency gains: "heavy investments in R&D," "more effort has to be put into developing technology that will reduce energy usage and emissions," and "technology exists to reduce global warming, but

our leaders have to get together to do it.” In some sense, hope now becomes linked to a belief that someone will do something, but a recent global survey found that a majority of those surveyed said that institutions, including businesses, NGOs, governments, and the media, are doing poorly in addressing existential challenges such as the recent pandemic and climate change.

Growing up, the extinction I confronted as a child was the result of rapid technological progress, possibly too rapid and too uncontrolled by social conventions, but today’s extinction seems more related to policy failures and the inability to mobilize technological progress, political will, and collective action rapidly enough, in time to head off future calamity. Given the mounting stakes, I thought there must be someone in charge of ‘extinction prevention’ so I started looking.

NASA’s headquarters is located in a complex of federal buildings in southwest Washington. Lindley Johnson, a retired Air Force Lieutenant Colonel, heads up the Planetary Defense Coordination Office (PDCO). As he describes it, “our mission is to find near-Earth objects (NEOs), warn of potential impacts with Earth, work on plans to mitigate potential impacts, and coordinate with other federal agencies and also foreign space agencies on planning for planetary defense.”⁸³ The results of this work became apparent to a global audience on September 22, 2022, when, after a 10-month space flight, NASA’s Double Asteroid Redirection Test (DART) – the world’s first planetary defense technology demonstration – successfully impacted a small asteroid moonlet called *Dimorphos* to nudge it out of its normal path. The 1,260-pound (750 kilogram) DART spacecraft traveled 7 million miles (11 million kilometers) to hit a target just 530 feet (150 meters) in diameter and we Earthlings actually got to see the impact.⁸⁴ Eight years before the DART initiative, NASA had actually assembled groups of citizens, using a technique known as participatory technology assessment (pTA), to discuss the proposed asteroid mission and discovered that they independently supported a federal role in protecting the Earth from asteroid collisions.⁸⁵ The public likes NASA according to recent surveys, with three quarters of Americans voicing a positive opinion of the agency.⁸⁶ Casual tourists in many foreign cities are bound to run into kids wearing NASA-branded gear -- t-shirts, hats, sweatshirts, or tote bags -- kids who would not be caught dead with an FDA or EPA logo across their chests or backs (in 2021, NASA fielded over 10,000 requests for permission to use their logo).⁸⁷ But the public has far different priorities for NASA than the agency, and, interestingly, protecting our planet from Asteroid strikes tops the list (60 percent), followed closely by “monitor[ing] key parts of the Earth’s climate system” (50 percent).

NASA also has a *Planetary Protection Officer*, Nick Benardini, who holds a Ph.D. in microbiology, molecular biology and biochemistry. Johnson likes to say that “he deals with that big, hard stuff and planetary protection deals with the small, squishy stuff.” Planetary protection actually has a much longer history going back to the before the Apollo moon missions. The 1967 United Nations Outer Space Treaty required that any exploration of the moon or other celestial bodies be conducted so as to, “...avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose.”⁸⁸ The Office of Planetary Protection was designed to protect other life forms in the universe from us — from our microbes, viruses, and other contagions — as well as ensuring

that astronauts did not bring back some alien life form from outer space that could wipe out humanity once the spacecraft was opened. These scenarios are termed 'forward' and 'backward' contamination and were a significant fear as NASA turned its sights on travel to the moon.

A 1962 study by the National Academy of Sciences warned that exploratory space missions risked, "the introduction into Earth's biosphere of destructive alien organisms," which could cause, "a disaster of enormous significance to mankind."⁸⁹ As NASA pushed forward through the 1960s towards landing men on the moon's surface, pressure increased to address backward contamination, resulting in millions being invested in the creation of a massive 86,000 square foot Lunar Receiving Laboratory in Houston, Texas, and the development of biological containment measures to deal with the ocean landing and transportation of the returning astronauts and their lunar cargo. John Rummel, who served for two years as NASA's planetary protection officer noted that, "If lunar organisms capable of reproducing in the Earth's Ocean had been present, we would have been toast."⁹⁰

These potential risks were amplified by two coincidental and publicly visible events that heightened anxieties of possible contamination if Apollo 11 brought back potentially dangerous lunar microorganisms. The first was the outbreak of the so-called Hong Kong flu which affected all 50 states by December 1968, ultimately claiming around 100,000 lives in the US and one million globally. *The New York Times* described it as "one of the worst in the nation's history." The second was the publication of Michael Crichton's 1969 novel *The Andromeda Strain*, which described, in chilling detail, a scenario similar to what faced NASA planners: a deadly extraterrestrial microbe, code named Andromeda, returned to Earth by a satellite. The press amplified public anxiety, for instance, with a feature in the popular *Saturday Evening Post* that stated, "if moon life does exist...it would be hostile to earth life, and...so different from anything on earth that neither plant, animal, nor human life would have any resistance against it."⁹¹

Given the complexity of the mission and short time frame need to prepare, NASA could not solve all the potential challenges involved in protecting the biosphere. The US was under enormous pressure to beat the Russians to the moon and maintaining the health and safety of the astronaut took priority. President Nixon was part of a memorable photo op that captured the beaming faces of astronauts Neil Armstrong, Michael Collins and Buzz Aldrin after their return. How effective were NASA's countermeasures? In the end, not very. A recent study by Georgetown University science historian Dagomar Degroot indicated that NASA officials knew that lunar microbes could pose an existential threat to Earth's biosphere and that their contamination countermeasures could not eliminate the threat. In the final analysis, they simply lucked out and maybe that will be what ultimately saves us from planetary threats -- luck.⁹² With new plans to go to Mars and return samples of the Martian environment in 2033, the challenge of assessing potential risks has again appeared.

An asteroid strike or release of a foreign pathogen is a LP/HR (low probability/high risk) event and we are confronted with these types of events often as one observer noted: "...as with viral outbreaks, the likelihood of a catastrophe is unlikely in any given year, but almost inevitable over time."⁹³ "The tendency in scientific and technical endeavors to overlook low-

probability but potentially existential risks was not unique to the Apollo program. Scholars and journalists have revealed that federal bureaucracies in the United States have mismanaged nuclear weapons and deadly pathogens, often prioritizing expediency or geopolitical competition over public safety."⁹⁴ A rush to be first, whether to the moon or to the market, often leads to failures to adequately assess and manage risks by both public and private sector entities and leaves little time for public deliberation and transparent communication. This push to act – to do something -- often leads governments to respond to long-term risks with short-term solutions, an approach that Michael Lewis has termed the *fifth risk*.⁹⁵

It is important to know the speed of the oncoming threats, which helps estimate the time remaining for countermeasures; the number and types of species affected; and the geographic boundaries of the impacts: local, regional, and or global. For instance, geologists and climatologists often talk about the *sixth mass extinction*, which is defined as a loss of around 75 percent of all species on the planet over a geologically short interval (less than 3 million years).⁹⁶ But maybe we have only 30 years left, according to a recent report suggesting that there is a high likelihood of human civilization coming to an end starting in 2050.⁹⁷ That still provides us with some time to prepare and adapt, if we can act. But this time interval would be shortened significantly through an exchange of nuclear weapons, even at a small scale. UN Secretary-General Antonio Guterres recently warned that, "The world is one step from 'nuclear annihilation.'"⁹⁸ How long is one step? Maybe just 2-5 years. A regional war between India and Pakistan involving fifty 15-kiloton nuclear weapons (the size of the Hiroshima bomb) could result in over 40 million casualties and be "sufficient to produce the lowest temperatures Earth has experienced in the past 1,000 years - lower than the post medieval Little Ice Age."⁹⁹ Recent models of food security impacts have indicated that more than 2 billion people could die globally within 2-5 years from starvation from a nuclear exchange between India and Pakistan and over 5 billion from a war between the United States and Russia.¹⁰⁰ Maybe we will have no warning at all, as some have maintained in the case of super intelligent AI systems: "we might not even know when a super-intelligence beyond our control arrives."¹⁰¹ Or these complex systems, like climate change, may have tipping points or domino effects that rapidly accelerate negative impacts along multiple dimensions. For instance, long duration heat waves could trigger electric grid failures and a loss of air conditioning, significantly increasing heat related deaths and illness. A recent study indicated that such an occurrence in Phoenix, Arizona, which, in July 2023, experienced almost an entire month where daytime temperatures exceeded 110 degrees Fahrenheit, could result in half the city's population requiring medical attention.¹⁰²

Confronted with the enormity of these scenarios, people and their representative governments do sometime act. For instance, humanity did address the ozone hole, first confirmed by British scientists in 1985 and based on earlier research findings published by University of California/Irvine scientists Sherwood Rowland and Mario Molina in 1974. The expanding hole over Antarctica became highly visible through dramatic pictures taken by NASA, was tied to a malignant and common form of cancer (melanoma), and was linked to a limited class of chemicals (initially chlorofluorocarbons or CFCs) being produced by a few companies, which were able to eventually develop substitutes. The Montreal Protocol, passed in 1989, was ultimately ratified by 197 nations and the European Union, and is often used as an example of successful international cooperation to address a global threat. The time between the basic scientific research (1974), the identification and confirmation of a global threat (1985)

and the beginning of concerted action (1989), was extremely short. But such successes in addressing global threats have become ever rarer.¹⁰³

Mikhail Gorbachev, who was instrumental in initiating reductions in nuclear weapons starting in the mid-1980s, once remarked that, “Models made by Russian and American scientists showed that a nuclear war would result in a nuclear winter that would be extremely destructive to all life on earth; the knowledge of that was a great stimulus to us, to people of honor and morality, to act in that situation.”¹⁰⁴ But more often threats trigger little action and, even as the threats unfold and multiply, little introspection or intervention. The fictional President in the Netflix film *Don’t Look Up* summed it up: “Do you know how many the-world-is-ending meetings we have had over the last two years? Drought, famine, hole in the ozone. It is so boring.”

Some thirty years ago, Anthony Fauci, Director of the National Institute of Allergy and Infectious Diseases, described his worst nightmare: “...it would be an outbreak of a respiratory illness that’s brand new, that’s easily spread, has a high degree of morbidity and mortality, and jumps from the animal reservoir to a human.”¹⁰⁵ Since then, we have had not one, but two, of these scenarios unfold, caused by highly pathogenic coronaviruses that jumped from animals to the human population. In 2002, the severe acute respiratory syndrome coronavirus (SARS-CoV) appeared causing fatal respiratory illnesses in humans, followed in 2011 by the Middle East respiratory syndrome coronavirus (MERS-CoV). Then came the COVID global pandemic, infecting 700 million people and killing almost 7 million by mid-2023. A study done early in the pandemic covering 81 countries found that over 20.5 million years of life had been lost because of COVID-19 due to premature deaths, a number that has probably tripled.¹⁰⁶

As we are confronted with extinction scenarios it is important to understand the psychological calculus that humans apply to catastrophic losses, which has been summed up as, “We value individual lives greatly, but these lives lose their value when they become part of a larger crisis.”¹⁰⁷ Psychologists term this tendency to focus on one life the *singularity effect*. Social media is full of videos showing individuals who risk their own lives to save another person, or a dog, cat, deer, or other animal. The consequences of this bias are significant since the result is often that we “do too little to protect human lives in the face of catastrophic treats from violence, natural disasters, and other causes.” As catastrophic risks pile up — climate change, the possible use of nuclear weapons, AI threats, another global pandemic — the common response is numbness to massive loss.¹⁰⁸

In many of these cases, significant death and suffering could be avoided. We are facing what Max Bazerman at Harvard Business School has termed *predictable surprises*, “...an event or set of events that take an individual or group by surprise, despite prior awareness of all the information necessary to anticipate the events and their consequences.”¹⁰⁹ Some characteristics of predictable surprises are that leaders knew about the problem, knew it would not fix itself, knew it was worsening over time, and did nothing. How predictable are predictable surprises? In 2001 and again in 2013, the European Environmental Agency undertook a comprehensive study of a wide range of environmental and public health hazards called *Late Lessons from Early Warnings* to better understand why, when scientific evidence of risks to human health and the environment existed, no action, or delayed action, was taken by governments or corporations.¹¹⁰

Case studies included chemicals like DDT, lead in gasoline, vinyl chloride in plastics, and mercury in fish; as well as threats such as climate change, increased floods, nuclear accidents, and invasive species. Of the 88 cases examined, only four represented examples where governments overreacted to an emerging risk which never materialized as predicted -- such as the US swine flu scare in 1976 or the artificial sweetener, saccharin. In a majority of the cases, the warnings proved correct.

You would think that an organization with the motto “Be Prepared” would pay attention to early warnings and act proactively to prevent disasters. But 40,000 teenagers from 158 countries attending the World Scout Jamboree in South Korean were caught off guard by extreme heat that hospitalized over 130 scouts and required hundreds to be treated on-site for heat-related symptoms. A 2018 report had warned the organizers that, “There is a high possibility that the [jamboree] event in August can be hit by an extreme heat of 36C [96.8 degrees Fahrenheit] and a typhoon.”¹¹¹

Early warnings -- with little action, late action, or no action. Why? Action is often hampered by a small minority that benefit from inaction and lobby to make sure nothing happens to threaten their position and its benefits. This should sound familiar. Often fixing these problems would be costly in the short term, while benefits are delayed. This situation results in governments responding to long-term risks with short-term policies and interventions -- placing Band-Aids over gapping wounds -- an approach that Michael Lewis has termed the *fifth risk*.¹¹² There are also a host of psychological reasons why individuals adapt a wait-and-see attitude -- from egocentrism to a comfort with the status quo. When I asked Bazerman about why policy makers and nations fail to act in the face of large scale, global threats, he pointed to politics and political polarization that often undercut the ability to recognize, prioritize, and act on emerging risks. This state of affairs often combines with an unerring belief in the magic of free markets to fix problems without public sector interventions or trust in companies to police their own actions and take corrective action.¹¹³ Who should address the risks of artificial intelligence, for instance? According to people such as Google’s former CEO Eric Schmidt: “There’s no one in the government who can get it right...But the industry can roughly get it right.”¹¹⁴ One member of U.S. Congress, Ted Cruz, the republican Senator from Texas, agreed, emphasizing that the Congress “doesn’t know what the hell it is doing” regarding AI.¹¹⁵

What separates many of today’s risks from threats that humankind faced a century ago is their increasing interconnectedness. Many of today’s risks do not exist in isolation and the failure to address a predictable surprise in a timely and effective manner can lead to the emergence of lower probability events, less predictable, with larger potential impacts. For instance, a warming planet could increase the potential release of what some have termed ‘time travelling pathogens’, biological organisms that have been trapped in glaciers and the permafrost for centuries or millennia, a possibility first raised by researchers in 2004.¹¹⁶ Included in this soup of organisms are genes that are resistant to a wide range of today’s antibiotics as well as viruses with the potential to infect both plants and fungi, including some taken from 30,000 year old ice samples from the Siberian permafrost. Just how large this risk is to both humans and ecosystems remains under-researched, despite our access to better and more sophisticated analysis techniques, and researchers have argued that, ‘...studies of microorganisms in glaciers and permafrost could help us to take a glimpse into the future, by

understanding and anticipating the potential threats we will face when [they] are released from their confinement.”¹¹⁷

There may also be positive applications lurking in our past – the threats of harm are accompanied by the promises of help and salvation. Scientists at the University of Pennsylvania have used machine learning to search for novel antibiotic candidates in the DNA of our distant relatives, the Neanderthals, who disappeared over 40,000 years ago, and the Denisovans, who roamed the planet between 15,000 and 30,000 years ago (an approach termed *molecular de-extinction*).¹¹⁸

We have reached a point in time where for every good there now seems to be two bads, where events no longer cancel out or ultimately skew positive, guiding us to the fork in the road that leads to a happy ending. For most of us, it is becoming increasingly difficult to counter the constant tug of pessimism in the face of escalating threats, especially as these threats rise to the top of our social media feeds and remain there for months, if not years. Adding to this din are the voices of a growing group of doomsdayers that make their living peering into the void created by extinction events, wallowing in worse-case scenarios, preparing, no doubt, for a predictable surprise. They are the natural heirs and offspring of those in the 1950’s who stockpiled canned goods, built fallout shelters, and plotted escape routes from ground zero. Many of these people belong to the upper crust of society’s technological elites, experts who have been intimately involved in creating the technologies they now seem to fear; many live in California. Already at the dawn of the Internet Age, people were pointing out the contradictions of the so-called ‘California Ideology’, a “technological determinism...not simple optimistic and emancipatory, [but] simultaneously a deeply pessimistic and repressive vision of the future.”¹¹⁹ Mo Gawdat, an artificial intelligence expert and ex-chief business officer at Google X, has warned people who don’t already have children to hold off, saying that the, “The risks are so bad, in fact, that when considering all the other threats to humanity, you should hold off from having kids if you are yet to become a parent.”¹²⁰ Luckily my parents didn’t get that message about the risks of the nuclear apocalypse in the 1950’s.

But many of today’s dystopians plan for extinction, some welcome it, assuming they may be the few humans remaining who, with the surviving nematodes and cockroaches, take over control of planet, or what is left of it. They are the ones (chosen or lucky) who have isolated far enough away from an impending disaster, carry some genetic fluke that saves them from an errant virus, moved their assets into gold, downloaded their brains via neural transplants and jettisoned their fragile bodies, or bought an advance ticket on the first flight to colonize Mars – the fortunate ones.

Maybe the time has come for humanity to shut off the light switch and exit; to do the planet a favor -- extinction as an extreme form of cosmic cleansing. The Australian philosopher Patricia MacCormick takes an extreme position, lamenting that “I am deeply saddened that there has never managed to be an annihilation of the human species, in spite of plague and war.”¹²¹ Total extinction is hard to even imagine since it would result in the loss of 8 billion humans along with other species as well as the loss of almost all historical records developed by humanity. Traces of our existence would remain, for instance, the Voyager spacecrafts launched in 1977 carry 12-inch records containing sounds and images designed to reflect life and culture

on Earth.¹²² There would also be an ever-decreasing stream of electromagnetic signals, remnants of human communications over a short 120-year span from Marconi's first radio transmissions through AM and FM broadcasts, television, and microwave signals. Exactly who would inherit the earth after human departure remains speculative, but certainly bacteria have an excellent chance as well as any remaining primates, like baboons, who might be able to figure out how to use tools and machines left by humans and kick-start a new existence and some semblance of technological progress.¹²³

Faced with extinction what would you do? What would you expect from others if their behaviors had no lasting impact on a future world? The 15th century German theologian, Martin Luther, had an answer to this question: "Even if I knew that tomorrow the world would go to pieces, I would still plant my apple tree." Would you? That is a difficult proposition to test but an analysis of 270 million records of player behavior in a massively multiplayer on-line role-playing game (MMORPG) found that players faced with a simulated extinction scenario became both more asocial, engaging in violent behaviors, and pro-social.¹²⁴

But wait, there is salvation.

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The first comic to explore the promises and perils of atomic energy was *Atoman*, which appeared in 1946. It addressed important issues and raised questions such as "How will Atoman use his strength? Will it be for good or evil? "Atomic power is too dangerous to be controlled by one man or one corporation. All the people should benefit from it!" Atomic energy appeared in both good and evil guises in the late 1940's in a number of Captain Marvel comics, with Captain Marvel vanquishing Mr. Atom to a lead vault but stressing that "Mr. Atom is a menace that the world cannot safely ignore." See: Wright, Bradford. 2001. *Comic Book Nation: The Transformation of Youth Culture in America*, Baltimore, MD: The John Hopkins Press.

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This calculator estimates the internal and external radiation dose received by your thyroid gland from radionuclides in fallout from nuclear tests conducted at the Nevada Test Site (NTS) and sites outside of the United States (global fallout). In addition, the calculator estimates your risk of developing thyroid cancer from that exposure. This calculator also provides an estimate of probability of causation (PC), sometimes called assigned share (AS) for individuals who have been diagnosed with thyroid cancer.

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